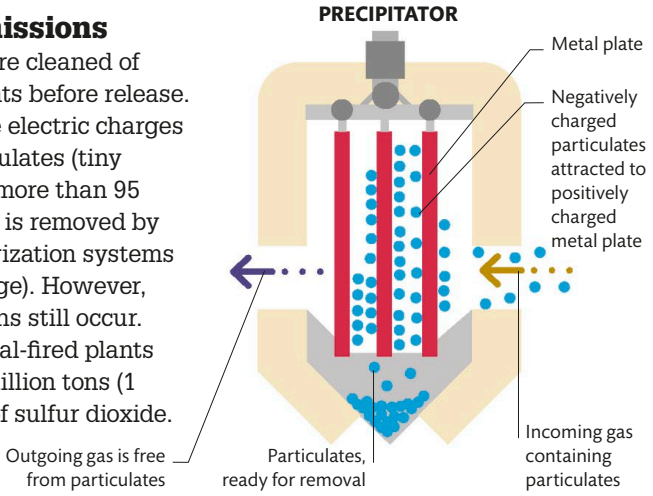
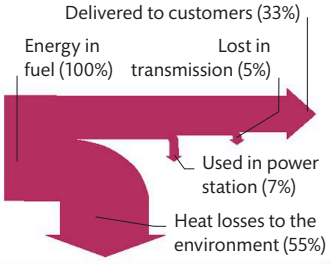
Cleaning emissions

Furnace gases are cleaned of harmful pollutants before release. Precipitators use electric charges to remove particulates (tiny particles) while more than 95 percent of sulfur is removed by flue gas desulfurization systems (see opposite page). However, harmful emissions still occur. Each year, US coal-fired plants emit about 1.1 million tons (1 million tonnes) of sulfur dioxide.



ENERGY EFFICIENCY

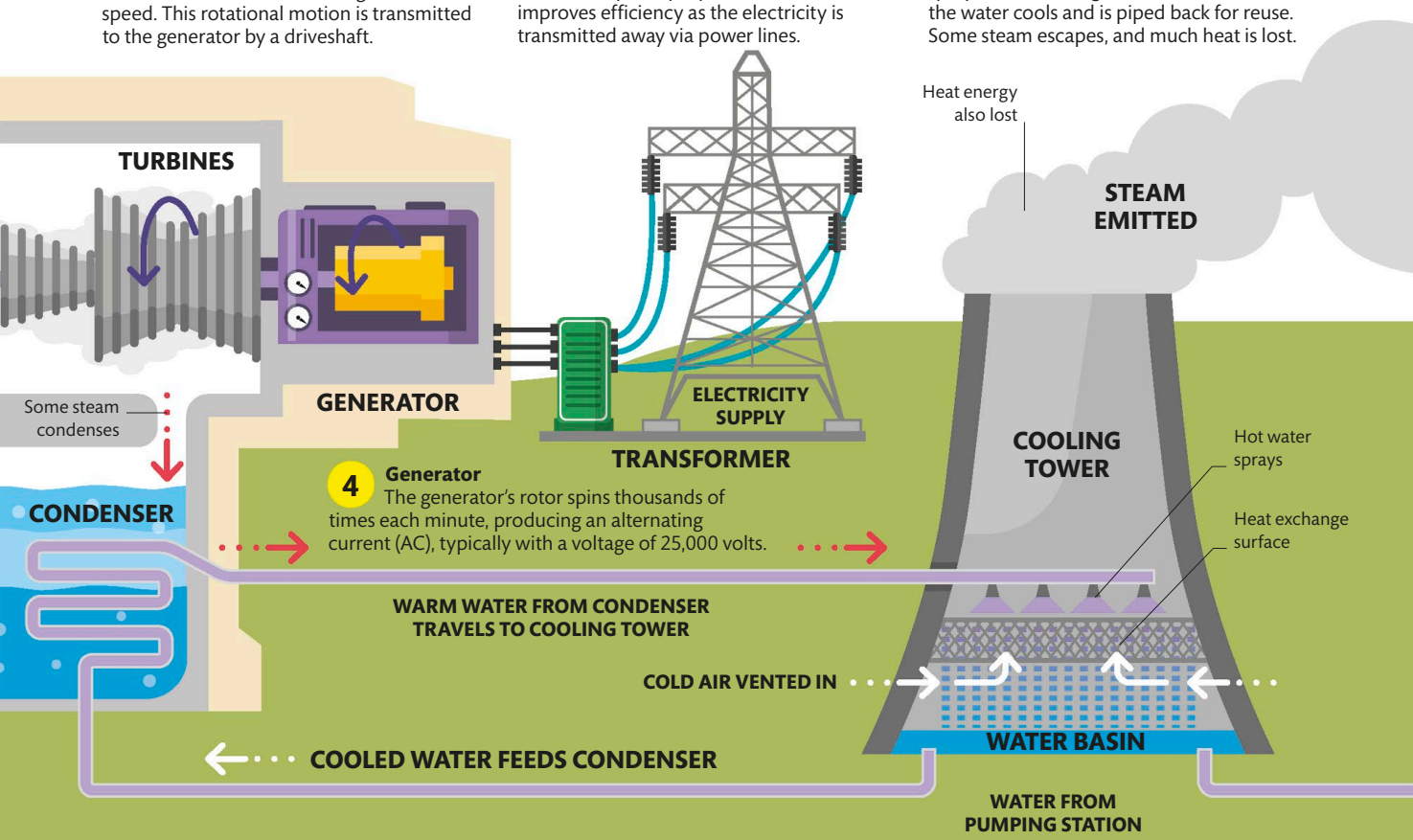
Only around one-third of all the energy in fuel reaches the user. More than 60 percent is lost at the power station.



3 Turbine
The high-pressure steam turns the fan blades of steam turbines with great force and speed. This rotational motion is transmitted to the generator by a driveshaft.

5 Electricity supply
The voltage of the electricity is greatly increased by a step-up transformer. This improves efficiency as the electricity is transmitted away via power lines.

6 Cooling tower
Steam is cooled in the condenser then sprayed into cooling towers, where most of the water cools and is piped back for reuse. Some steam escapes, and much heat is lost.



4 Generator
The generator's rotor spins thousands of times each minute, producing an alternating current (AC), typically with a voltage of 25,000 volts.